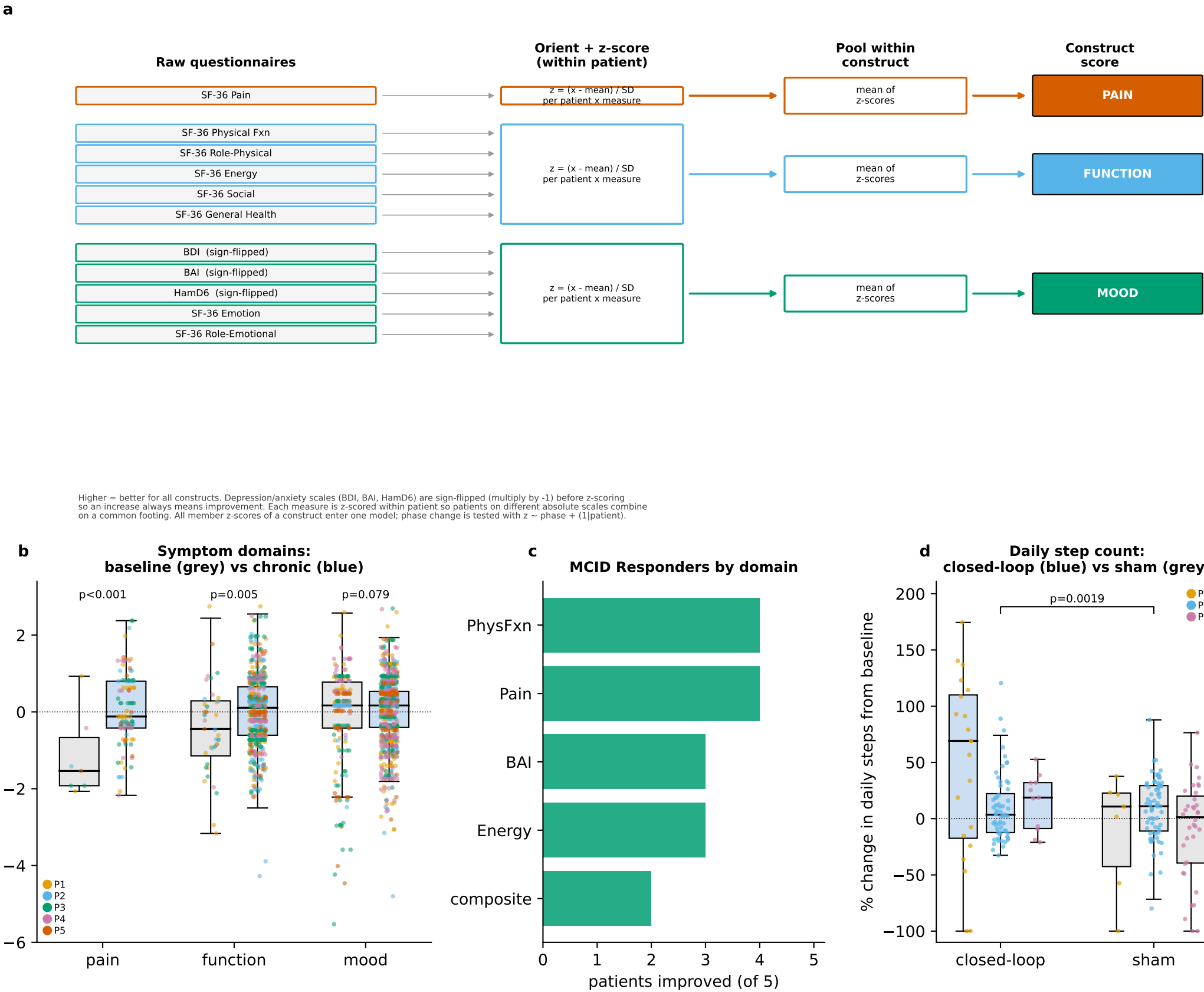

a
Raw questionnaires
Orient + z-score (within patient)
Pool within construct
Construct score
SF-36 Pain
SF-36 Physical Fxn
SF-36 Role-Physical
SF-36 Energy
SF-36 Social
SF-36 General Health
BDI (sign-flipped)
BAI (sign-flipped)
HamD6 (sign-flipped)
SF-36 Emotion
SF-36 Role-Emotional
z = (x - mean) / SD per patient x measure
z = (x - mean) / SD per patient x measure
z = (x - mean) / SD per patient x measure
mean of z-scores
mean of z-scores
mean of z-scores
PAIN
FUNCTION
MOOD
Higher = better for all constructs. Depression/anxiety scales (BDI, BAI, HamD6) are sign-flipped (multiply by -1) before z-scoring so an increase always means improvement. Each measure is z-scored within patient so patients on different absolute scales combine on a common footing. All member z-scores of a construct enter one model; phase change is tested with z ~ phase + (1|patient).
b
Symptom domains: baseline (grey) vs chronic (blue)
p<0.001
p=0.005
p=0.079
within-patient z (higher = better)
P1
P2
P3
P4
P5
pain
function
mood
c
MCID Responders by domain
PhysFxn
Pain
BAI
Energy
composite
patients improved (of 5)
d
Daily step count: closed-loop (blue) vs sham (grey)
p=0.0019
% change in daily steps from baseline
P1
P2
P4
closed-loop
sham